

Technical Note: FW402 Woven Geotextile in PT-ROW™ Pre-Treatment Applications

Overview

The PT-ROW™ system integrates AquaCell modular crates with engineered geotextile layers to enhance stormwater management performance. As part of this system, FW402 woven geotextile fabric (or approved equal) is used at the base of the crate stack to provide pre-treatment functionality prior to infiltration or detention.

Key Benefits of FW402 in Pre-Treatment

1. Sediment Filtration

FW402 acts as a coarse filtration layer, capturing larger sediment particles and debris before they enter the AquaCell system. This reduces the risk of internal clogging and extends the operational life of the system.

2. Hydraulic Performance

With a permeability of 0.14 cm/sec and a water flow rate of 145 gal/min/ft², FW402 allows efficient water movement while regulating flow velocity. This helps prevent scouring and promotes uniform distribution across the base layer.

3. Structural Separation

FW402 maintains separation between the subgrade and the aggregate base, preventing migration of fines into the stone layer. This preserves the porosity and infiltration capacity of the system over time.

4. Durability and Load Resistance

Constructed from woven slit-film polypropylene, FW402 offers high tensile strength and resistance to puncture, making it suitable for high-load environments such as parking lots and roadways.

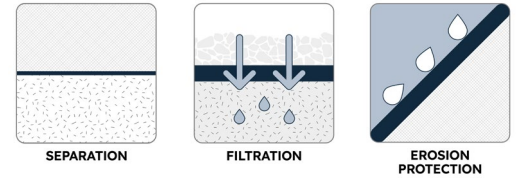
5. Pretreatment Zone Enhancement

By creating a passive filtration interface at the bottom of the PT-ROW™, FW402 enhances water quality before it enters the AquaCell crates. This mimics the function of a pre-treatment basin without requiring additional excavation or infrastructure.

Recommended Use

- Location: Bottom of AquaCell crate stack
- Function: Pre-treatment, sediment control, structural separation
- Installation: Wrap only the base layer; do not wrap the top to allow vertical infiltration

MIRAFI FW402



MIRAFI® FW402 is composed of high-tenacity monofilament polypropylene yarns, which are woven into a stable network such that the yarns retain their relative position. MIRAFI FW402 geotextile is inert to biological degradation and resists naturally encountered chemicals, alkalis, and acids.

TenCate Geosynthetics Americas (A Solmax Company) is accredited by Geosynthetic Accreditation Institute – Laboratory Accreditation Program ([GAI-LAP](#)).

MIRAFI FW402 meets Build America, Buy America Act, Pub. L. No. 117-58, div. G §§ 70901-52.

MECHANICAL PROPERTIES	TEST METHOD	UNIT	MINIMUM AVERAGE ROLL VALUE	
			MD	CD
Grab Tensile Strength	ASTM D4632	lbs (N)	365 (1624)	200 (890)
Grab Tensile Elongation	ASTM D4632	%	24	10
Trapezoid Tear Strength	ASTM D4533	lbs (N)	115 (512)	75 (334)
CBR Puncture Strength	ASTM D6241	lbs (N)	675 (3004)	
MECHANICAL PROPERTIES	TEST METHOD	UNIT	MINIMUM ROLL VALUE	
Percent Open Area	COE-02215	%	10	
Permittivity	ASTM D4491	sec ⁻¹	2.1	
Flow Rate	ASTM D4491	gal/min/ft ² (l/min/m ²)	145 (5907)	
			MAXIMUM OPENING SIZE	
Apparent Opening Size (AOS)	ASTM D4751	U.S. Sieve (mm)	40 (0.425)	
			MINIMUM TEST VALUE	
UV Resistance (at 500 Hours)	ASTM D4355	% strength retained	90	
PHYSICAL PROPERTIES		UNIT	TYPICAL ROLL VALUE	
Roll Dimensions (width x length)		ft (mm)	12.5 x 300 (3.8 x 91)	
Roll Area		yd ² (m ²)	417 (348)	
Roll Weight		lbs (kg)	178 (81)	

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